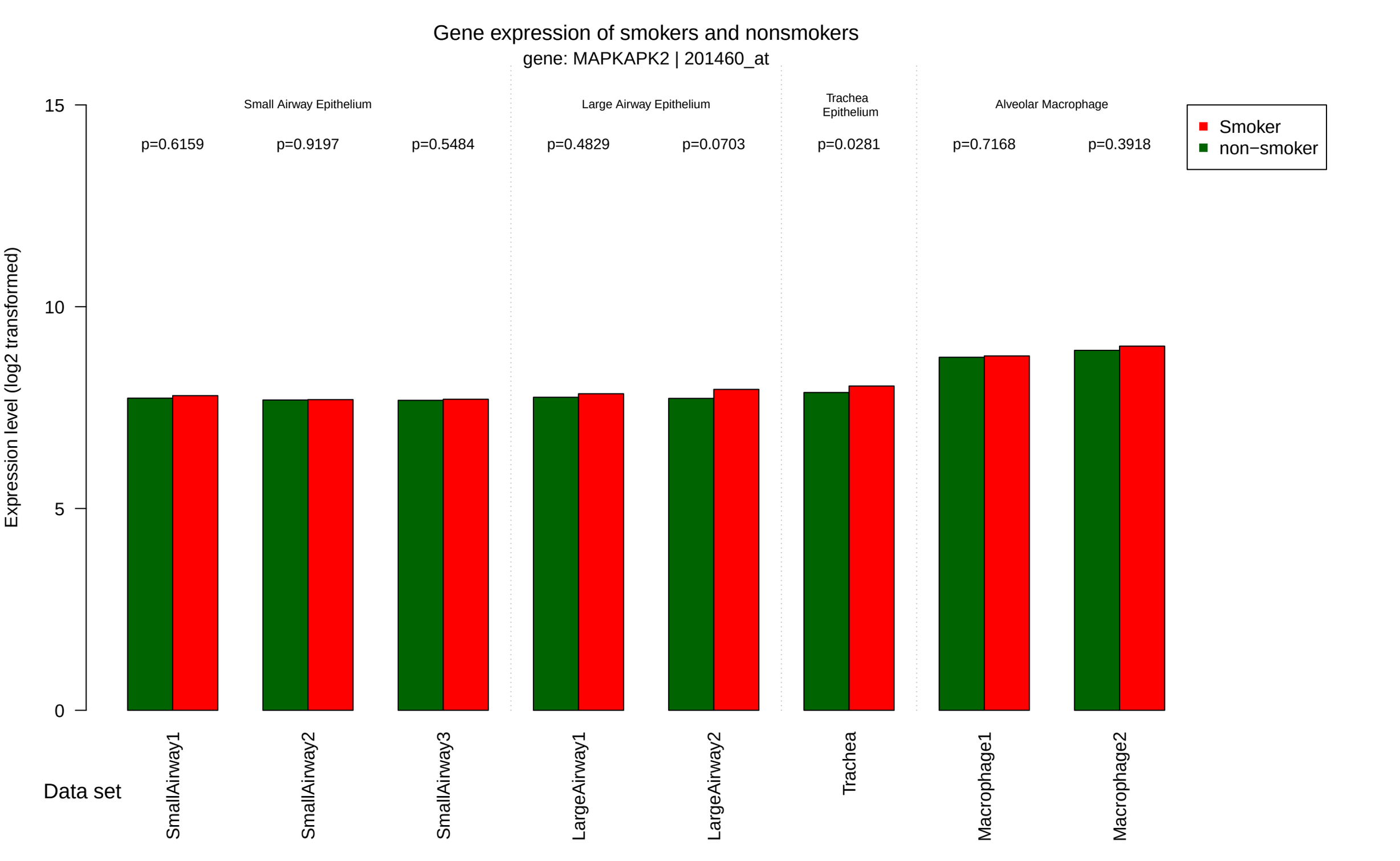
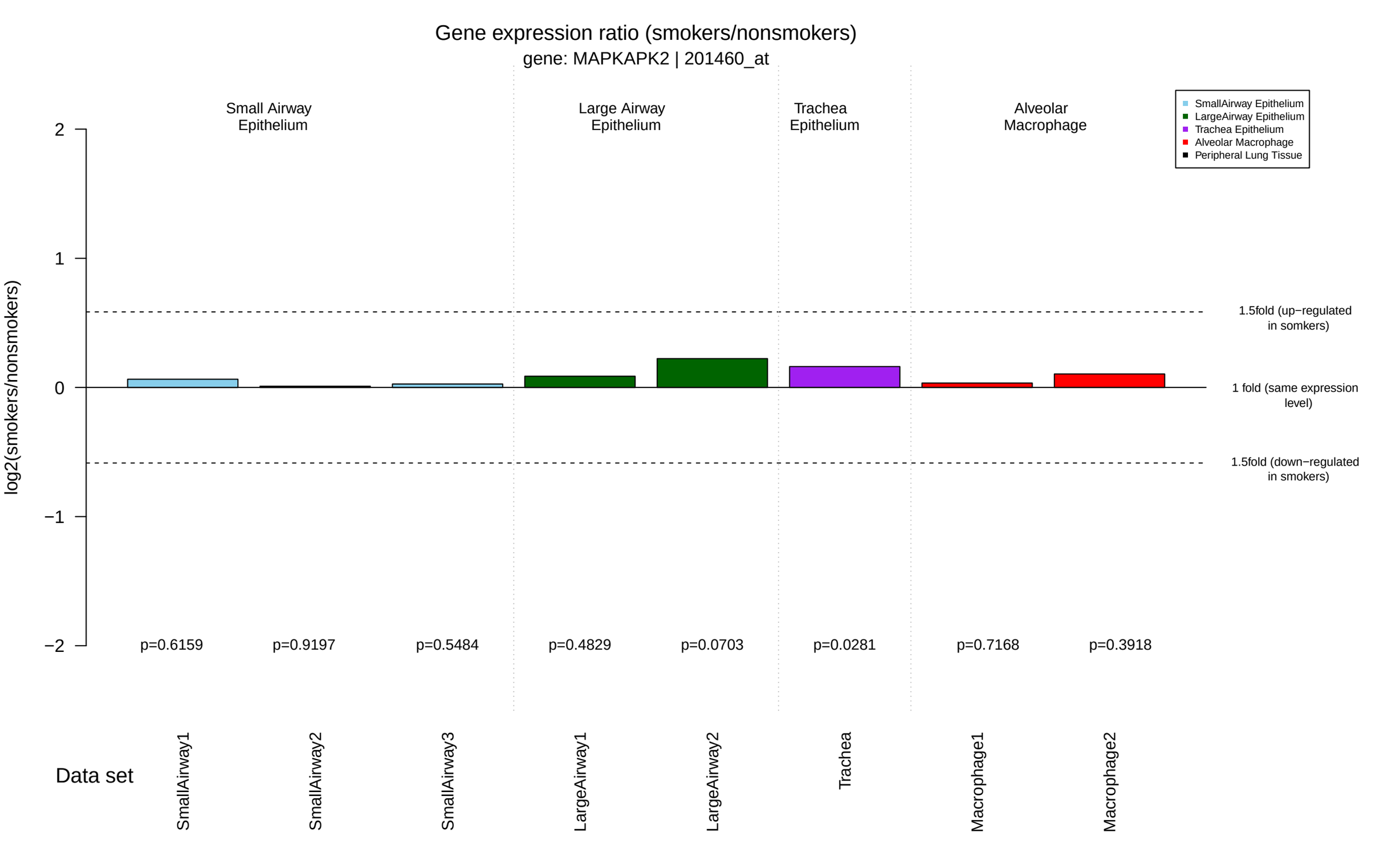
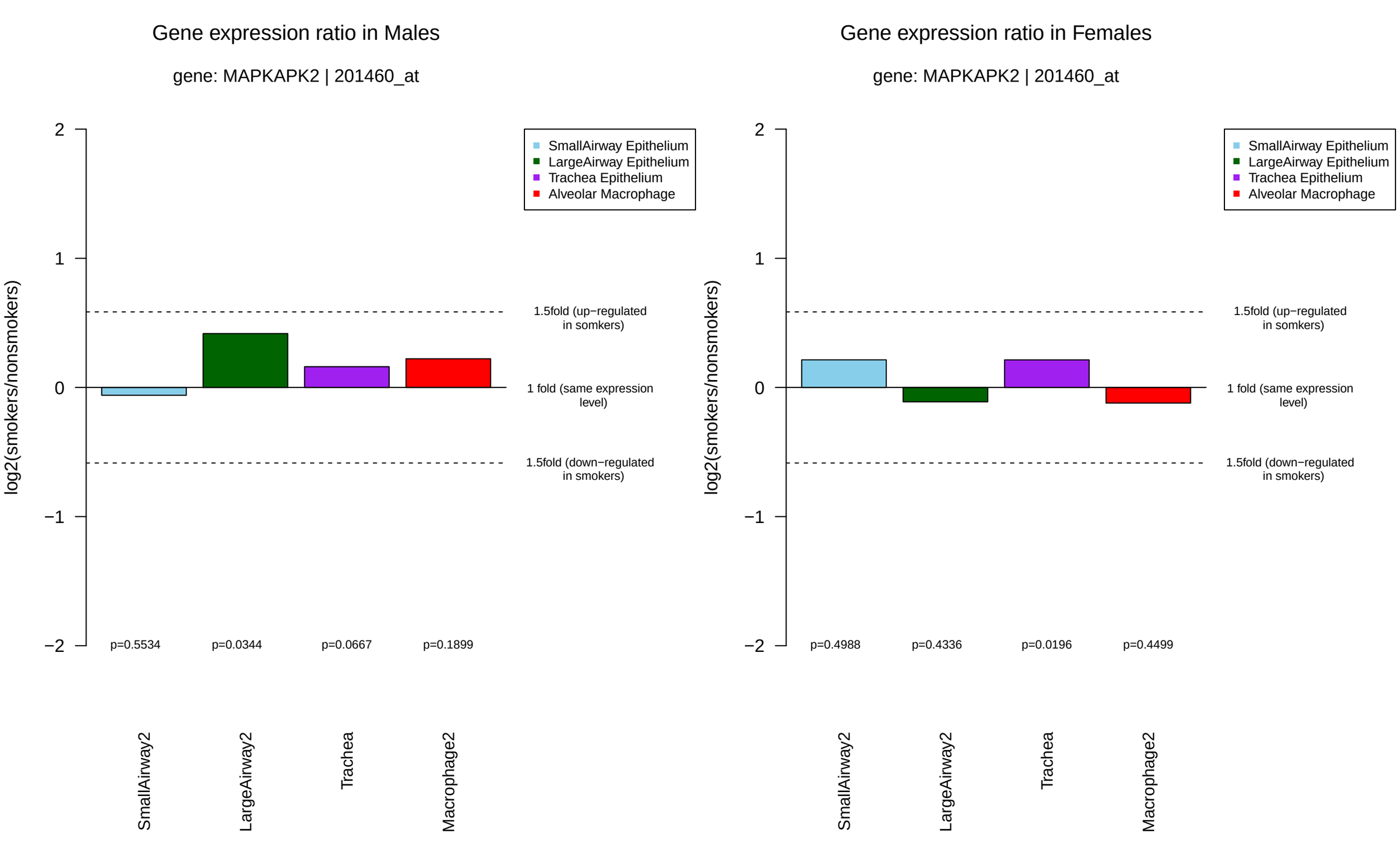
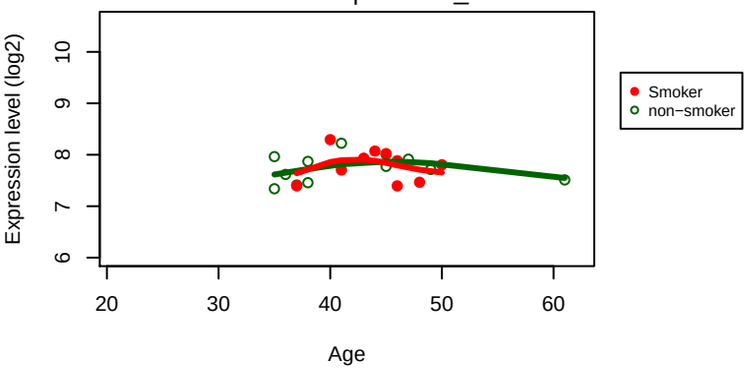
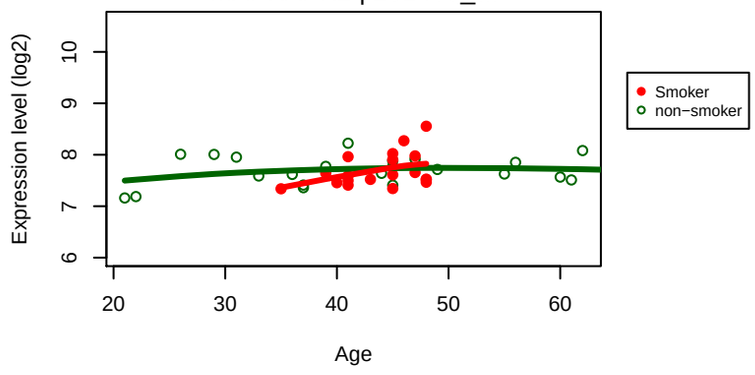
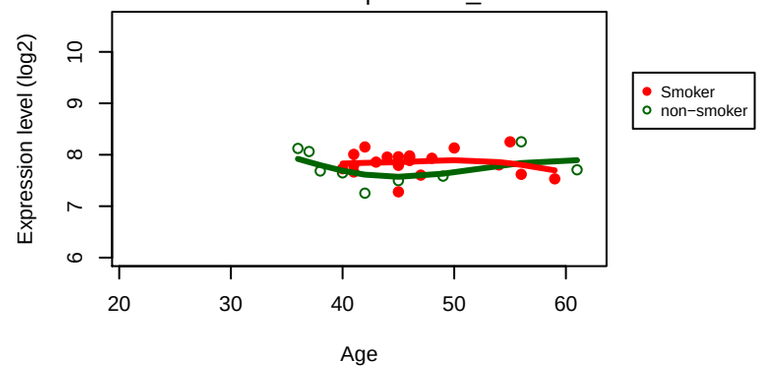

Gene expression of smokers and nonsmokers
gene: MAPKAPK2 | 201460_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6159
p=0.9197
p=0.5484
p=0.4829
p=0.0703
p=0.0281
p=0.7168
p=0.3918
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

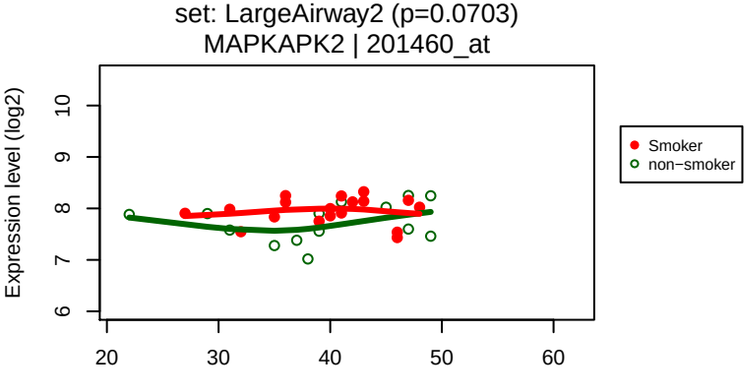

Gene expression ratio (smokers/nonsmokers)
gene: MAPKAPK2 | 201460_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6159
p=0.9197
p=0.5484
p=0.4829
p=0.0703
p=0.0281
p=0.7168
p=0.3918
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

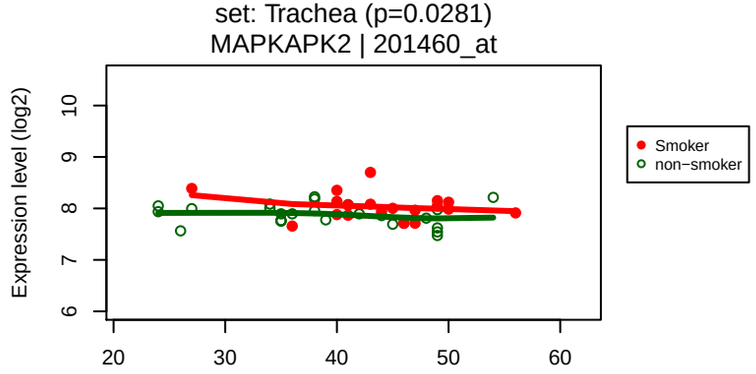

Gene expression ratio in Males
gene: MAPKAPK2 | 201460_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5534
p=0.0344
p=0.0667
p=0.1899
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPKAPK2 | 201460_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4988
p=0.4336
p=0.0196
p=0.4499
SmallAirway2
LargeAirway2
Trachea
Macrophage2

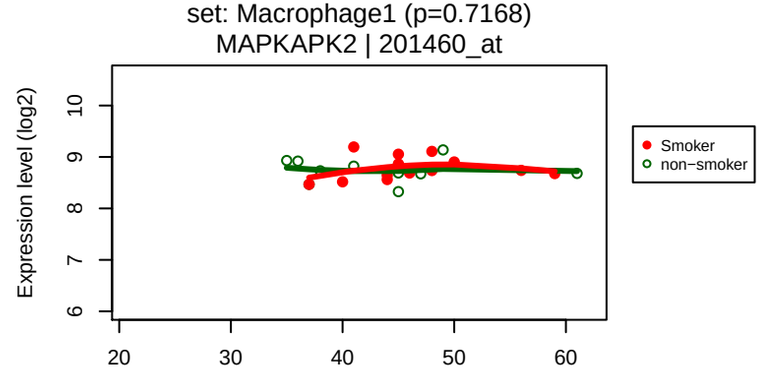

set: SmallAirway1 (p=0.6159)
MAPKAPK2 | 201460_at
Expression level (log2)
Age
Smoker
non−smoker

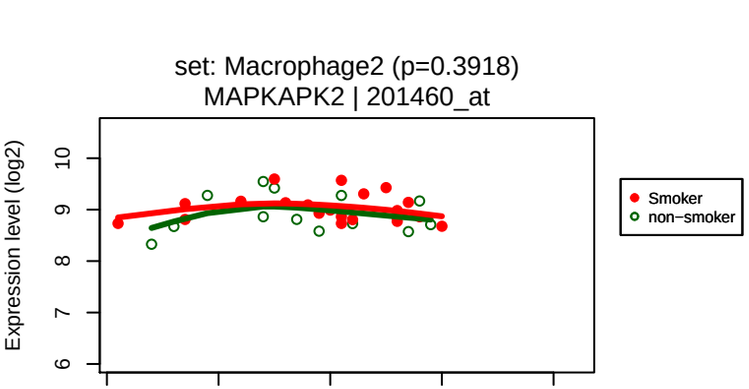

set: SmallAirway2 (p=0.9197)
MAPKAPK2 | 201460_at
Expression level (log2)
Age
Smoker
non−smoker

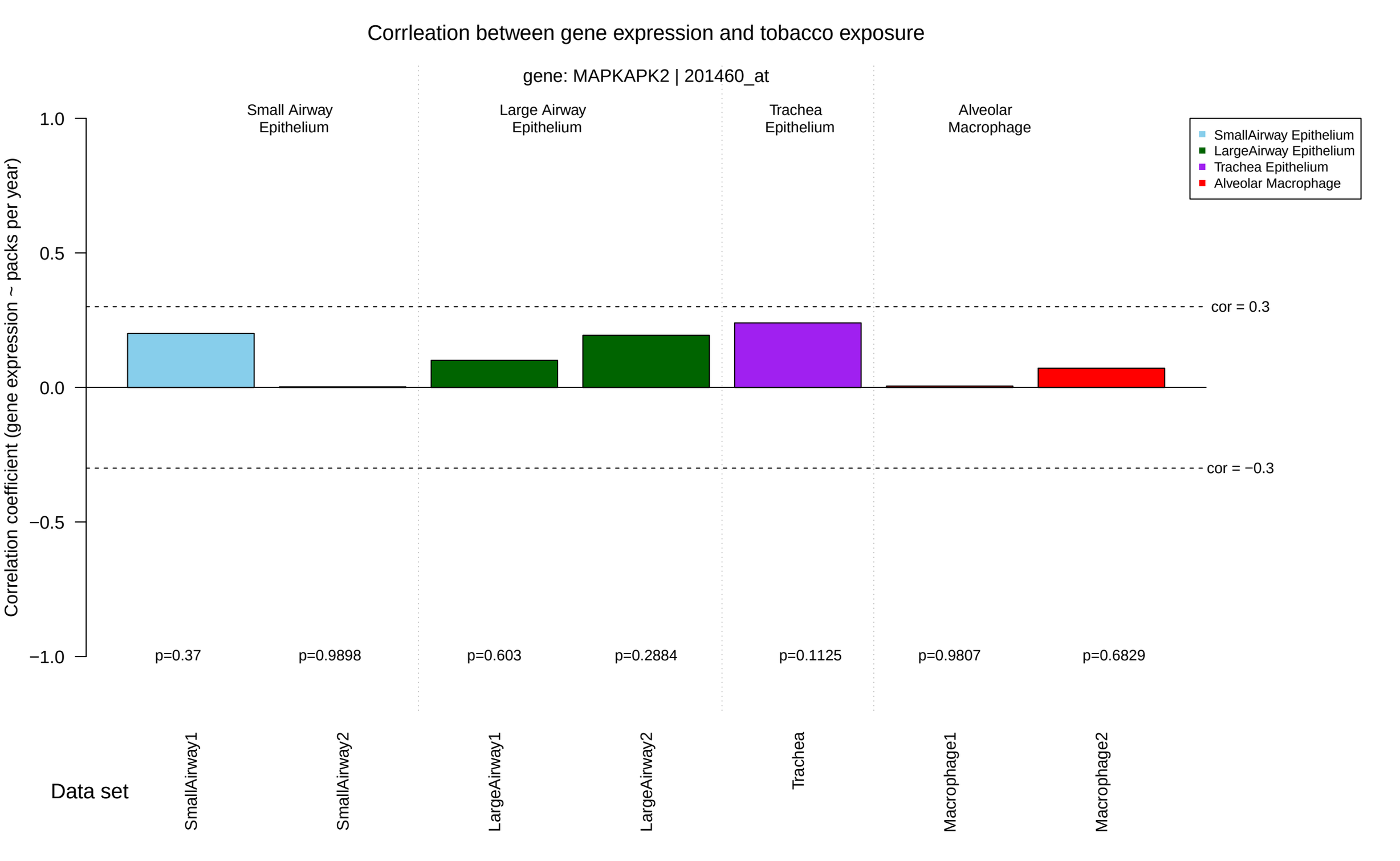

set: LargeAirway1 (p=0.4829)
MAPKAPK2 | 201460_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0703)
MAPKAPK2 | 201460_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0281)
MAPKAPK2 | 201460_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7168)
MAPKAPK2 | 201460_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3918)
MAPKAPK2 | 201460_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPKAPK2 | 201460_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.37
p=0.9898
p=0.603
p=0.2884
p=0.1125
p=0.9807
p=0.6829
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2